Machine Learning and Survival Models on Pediatric Brain Tumor EHRs

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Abstract—
This project presents an analysis of electronic health records of pediatric patients with brain tumors, aiming to identify the key factors influencing survival. The main goal was to understand which clinical, pathological, and treatment-related variables most significantly affect patient outcomes. The results highlight the critical role of factors such as extent of disease and type of surgery in determining survival. While machine learning models offered moderate predictive power, survival analysis provided valuable insights for prognosis. The study underscores the complexity of pediatric brain tumors and the importance of integrating diverse data sources for improved risk stratification.

Keywords—*Pediatric brain tumors, survival analysis, machine learning, clustering, Cox model*

1. Introduction

Pediatric brain tumors represent one of the most frequent forms of solid cancer in children and a leading cause of cancer-related mortality in the pediatric population. These tumors are highly heterogeneous in terms of histology, anatomical location, and clinical behavior, posing significant challenges for diagnosis, treatment, and prognosis. Despite advances in neurosurgery, radiotherapy, and chemotherapy, survival rates vary greatly depending on the extent of disease, tumor type, and available resources, particularly in middle-income countries.

Recent studies have emphasized the role of factors such as extent of disease, tumor location, and type of surgery in determining survival outcomes [1]. In this project, we conducted a comprehensive analysis of electronic health records (EHRs) from a national referral center. Our aim was to identify clinical, surgical, and pathological factors that influence survival in pediatric patients with brain tumors. To achieve this we perform e full pipeline, including data cleaning and preparation, exploratory data analysis (EDA), clustering to explore latent patterns, supervised machine learning models to predict outcomes, and survival analysis using both Kaplan-Meier and Cox proportional hazards models.

The insights gained from this study can contribute to better risk stratification, inform clinical decision-making, and highlight the potential of computational approaches in understanding complex oncological data.

2. Dataset Presentation

The dataset analyzed in this project consists of electronic health records (EHRs) from pediatric patients diagnosed with primary brain tumors and treated at the Institute of Oncology and Radiology of Serbia, the national referral center, over a ten-year period (2007-2016). The data comprise 174 records, corresponding to individual patients included in a retrospective cohort study. The study was designed to evaluate clinical, pathological, and treatment-related factors associated with survival outcomes in this population.

The dataset includes detailed information on demographics, clinical presentation, tumor characteristics, treatments, and outcomes, providing a comprehensive basis for analysis.

Demographic Variables

The demographic data comprise year of birth and gender of each patient. These variables provide basic context for understanding the distribution of cases within the cohort.

Clinical Variables

The dataset includes information on several clinical parameters:

- Presenting symptoms, such as increased intracranial pressure, neurological deficits, epileptic seizures, and hormonal abnormalities.
- Tumor location, categorized as supratentorial, infratentorial, or involving both compartments.
- Extent of disease at diagnosis, classified as no evidence of disease (NED), local residual disease (LRD), or disseminated disease (DD).

Pathological Variables

The histopathological type and grade of tumor are recorded for each case, based on available diagnostic reports. The classification follows the World Health Organization (WHO) criteria and includes various tumor types such as embryonal tumors, gliomas, ependymomas, germ cell tumors, craniopharyngiomas, and cases with unknown histology.

Treatment Variables

Treatment-related variables detail:

- Type of surgical intervention, distinguishing between gross total resection (GTR), subtotal resection, tumor reduction, and biopsy.
- Radiotherapy, indicating whether craniospinal irradiation or local radiotherapy was administered.
- Chemotherapy, specifying whether it was given as neoadjuvant, concomitant, or adjuvant therapy.

Outcome Variable

The primary target variable in this study is *survival status*, indicating whether the patient was alive or deceased at the last follow-up. The dataset also includes the corresponding *observation time*, defined as the time from diagnosis to death or last follow-up, which represents the overall survival (OS) period for each patient.

3. Data Cleaning and Preparation

The dataset was initially imported from an Excel file containing two sheets: the first with the data used for analysis, and the second providing variable descriptions.

Initial Structure and Formatting

To ensure consistency and code compatibility, all column names were reformatted by replacing spaces and special characters with underscores. Duplicates were checked and not detected. One observation with missing target values and extensive null fields was removed, reducing the dataset from 174 to 173 records.

Variable Renaming and Recoding

Several transformations were performed to enhance readability and prepare the data for analysis:

- The column `Age` was renamed to `Year_of_birth` to reflect its content more accurately.
- The variable `Sex` was recoded as a binary variable (1 = male, 0 = female).
- Binary categorical variables with values `1 = yes` and `2 = no` (e.g., `Increased_ICP`, `Epileptic_seizures`) were harmonized into standard 0/1 coding.

Surgical Variables

The `Operation_type` variable, which contains missing values for non-operated patients, was augmented with a new category (0 = no surgery). An additional binary variable `Operation_type_bin` was created to distinguish patients who underwent gross total resection (value 1) from all other cases (value 0).

Histological Variables

Tumor histology variables (e.g., `Embryonal_tumors`, `HGG`, `Ependymoma`) were originally encoded as 1 (yes) and 2 (no). These were standardized to binary 0/1 format. Each histological type was handled independently, allowing for further feature selection during modeling.

Extent and Imaging Variables

The variable `MRI_finding`, which had three missing values, was imputed using the correlated variable `Extent_of_disease`, preserving internal consistency. A high correlation between the two variables supported this decision.

CSF Cytology Integration

The information from `CSF_cytology` and `CSF_cytology_finding` was combined into a single binary variable named `CSF_cytology_combined`, which indicates the presence of malignant cells only when cytological examination was performed and returned a positive result.

Radiotherapy and Doses

Missing values in radiotherapy-related dose variables (`CSI_dose`, `Boost_dose`, `Total_dose`) were treated based on clinical logic. For patients who did not receive radiotherapy, missing doses were replaced with zero. In cases of partial data with known radiotherapy status, missing doses were also imputed as zero, assuming localized treatments.

Chemotherapy Variables

Variables indicating whether neoadjuvant, adjuvant, or concomitant chemotherapy was administered were recoded to binary format. Special attention was paid to `Concomitant_HT`, which contained invalid values and missing entries. Implausible values were set to `NA`, and missing values were imputed conditionally:

- If a patient did not receive radiotherapy, concomitant chemotherapy was set to 0.
- Remaining missing values were imputed using the `mice` package with logistic regression.

Target Variable Definition

The original variable `Status_OS` was used to derive the binary target variable `Death`, where 1 indicates death and 0 indicates survival at last follow-up. The observation time (in months) was maintained in the `OS` column and used later for survival analysis.

At the end of preprocessing, all variables were validated for consistency, and missing values were either addressed through logical imputation or handled via statistical methods.

4. Exploratory Data Analysis

An exploratory analysis was conducted to understand the clinical characteristics of the cohort and to investigate associations between key variables and survival outcomes. The primary target variable, `Death`, indicating whether the patient was deceased at last follow-up, was relatively balanced, with 79 deaths and 94 survivals. The gender distribution was also nearly equal, with 90 males and 83 females.

The overall survival time (`OS`), used as a continuous indicator of prognosis, displayed a right-skewed distribution with a peak in early mortality during the first 20 months and a long tail reflecting a subset of long-term survivors. This pattern is consistent with the heterogeneous nature of pediatric brain tumors.

Histological subtype emerged as one of the most relevant prognostic factors. High mortality rates were observed in patients with high-grade gliomas (HGG) and tumors of unknown histology, likely corresponding to radiologically diagnosed brainstem gliomas. Conversely, low-grade gliomas (LGG), craniopharyngiomas, and germ cell tumors were mostly associated with survival.

Tumor location further negatively influenced the prognosis. Patients with supratentorial tumors showed the best survival, followed by those with infratentorial lesions. Tumors involving both compartments were associated with the poorest outcomes, likely reflecting greater disease complexity and reduced resectability.

Surgical radicality was also a strong determinant of survival. Gross total resection (GTR) was consistently associated with longer OS and lower mortality, particularly among patients with localized disease (extent 1). However, in patients with disseminated disease (extent 3), the survival benefit of GTR was substantially reduced, suggesting that surgical efficacy is limited in the presence of systemic progression.

The extent of disease at diagnosis showed a clear prognostic gradient. Patients with no evidence of disease had the most favorable outcomes, while those with local residual or disseminated disease experienced progressively higher mortality. This observation confirms the central role of initial staging in predicting prognosis.

Radiotherapy was associated with slightly better survival on average, although outcomes were highly variable. Since radiotherapy is typically administered in more aggressive or advanced cases, its prognostic interpretation is confounded by indication bias. Similarly, total radiotherapy dose showed only a weak association with survival, further supporting the idea that treatment intensity often reflects disease severity.

The duration of symptoms before diagnosis showed a mild positive correlation with OS, possibly indicating that more indolent tumors lead to later diagnosis but better long-term outcomes. However, this relationship should be interpreted cautiously due to the influence of outliers and heterogeneous clinical presentations.

Finally, sex did not appear to influence mortality, with similar death rates observed among male and female patients. This finding supports previous evidence suggesting that gender is not a significant prognostic factor in this population.

5. Feature Engineering and Variable Selection

Prior to modeling, a structured phase of feature engineering and selection was conducted to ensure consistency, reduce redundancy, and improve the interpretability and predictive potential of the dataset.

Demographic and clinical variables such as sex, year of birth, neurological symptoms, and tumor dissemination were retained with appropriate encoding. Categorical variables like `Extent_of_disease` and `Localisation` were transformed into dummy variables through one-hot encoding, excluding baseline levels to avoid redundancy and ensure model stability.

Tumor histotypes, initially encoded as mutually exclusive binary indicators, were reviewed for frequency. Subtypes with fewer than 10 observations—such as craniopharyngiomas, pineal tumors, choroid plexus tumors, glioneuronal tumors, and unspecified types—were aggregated into a single binary feature named `Rare_histologies`. The resulting set of histotype variables included: `Embryonal_tumors`, `HGG`, `LGG`, `Ependymoma`, `GCT/NGCT`, `Unknown`, and `Rare_histologies`.

Among staging-related variables, `Extent_of_disease` was preferred over `MRI_finding` due to its more comprehensive coverage, incorporating both radiological and cytological data. Their strong correlation ($r = 0.86$) confirmed functional redundancy, leading to the exclusion of `MRI_finding`.

Treatment-related features such as chemotherapy indicators (`Neoadjuvant_HT`, `Concomitant_HT`, `Adjuvant_HT`) and radiotherapy dose variables (`CSI_dose`, `Boost_dose`, `Total_dose`) were initially preserved.

To assess redundancy and guide dimensionality reduction, a correlation matrix was computed. The most relevant associations are summarized below:

- `Total_dose` and `Radiotherapy`: strong positive correlation (r = 0.91), as patients who did not receive radiotherapy had zero dose.
- `CSI_dose` and `Boost_dose`: high correlation (r = 0.88), reflecting typical therapeutic protocols where the two are administered together.
- `Extent_of_disease2` and `Operation_type_bin`: strong negative correlation (r = - 0.89), suggesting that local residual disease is rarely associated with gross total resection.
- `Embryonal_tumors` and `CSI_dose` (r = 0.68), `Boost_dose` (r = 0.62), `Adjuvant_HT` (r = 0.55): all positively associated, reflecting aggressive treatment plans.
- `HGG` and `Concomitant_HT`: moderately strong positive correlation (r = 0.63), consistent with standard concurrent chemoradiotherapy.
- `GCT/NGCT` and `Hormonal_abnormalities`: correlation (r = 0.51), supporting the frequent sellar involvement of germ cell tumors.
- `Neoadjuvant_HT` and `Total_dose/Radiotherapy`: negative correlations (r = - 0.49), possibly indicating an alternative treatment pathway in select subgroups.
- `Embryonal_tumors` and `Localisation2`: correlation (r = 0.53), reflecting common infratentorial origin (e.g., medulloblastomas).

To further address multicollinearity, variance inflation factors (VIF) were computed using logistic regression with `Death` as the dependent variable. Given the extremely high correlation between `Radiotherapy` and `Total_dose`, both were tested in univariate models. As their AIC values were nearly identical, `Radiotherapy` was retained for simplicity and interpretability, and `Total_dose`, `CSI_dose`, and `Boost_dose` were excluded.

Additional removals included `Extent_of_disease2` and `Embryonal_tumors`, both highly collinear with other retained features. After these refinements, the final dataset was confirmed to be free of multicollinearity.

These findings were used in subsequent modeling stages. For clustering tasks, correlated variables were retained when contributing complementary information. In contrast, for regression and predictive modeling, multicollinearity was formally addressed using regularization techniques and variance inflation analysis.

6. Dimensionality Reduction and Clustering

To explore the intrinsic structure of the dataset and assess whether patients naturally group into clinically meaningful subtypes, we applied a sequence of dimensionality reduction and clustering techniques. Target variables `Death` and `OS` were excluded prior to reduction to avoid outcome leakage.

Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was conducted on the scaled dataset. The first principal component (PC1) accounted for approximately 18% of the total variance, followed by PC2 (10.7%) and PC3 (8.6%). No single component dominated the variance, indicating that the data are inherently high-dimensional. The projection onto the first two components explained less than 30% of the total variance, reflecting a complex multivariate structure without clear dominant axes.

UMAP and t-SNE Projections

UMAP was used to embed the dataset in a lower-dimensional space while preserving local structure. The resulting two-dimensional representation revealed some separation with respect to `Extent_of_disease`, as confirmed by significant t-tests on both UMAP dimensions comparing NED (extent 1) to others ($p < 0.001$). However, no clear separation emerged with respect to the `Death` variable, suggesting that survival status is influenced by multiple interdependent factors not linearly separable in low-dimensional space.

Similarly, t-distributed Stochastic Neighbor Embedding (t-SNE) did not yield visibly distinct clusters by survival outcome. The resulting plot showed significant overlap among groups, reinforcing the idea that mortality is not explained by simple variable patterns in the dataset.

K-means and Hierarchical Clustering

We applied K-means clustering to both the raw scaled data and the UMAP projection for k values ranging from 2 to 5. On the raw dataset, clustering yielded poor separation (average silhouette < 0.25, Davies-Bouldin index > 1.6), with no clear elbow in WSS. This indicated an absence of strong cluster structure in the original feature space. Hierarchical clustering using Ward's method on the raw dataset failed to produce strongly defined groups. Average silhouette scores were low, and Davies-Bouldin index remained above 1.6. Dendrograms showed overlapping branches with no clear division.

In contrast, clustering in the UMAP space revealed improved cohesion and separation. The silhouette score was 0.55 and the Davies-Bouldin index dropped below 0.7 for k = 3, supporting the presence of a more stable and interpretable cluster configuration. K-means on UMAP with k = 3 yielded clinically interpretable clusters:

- *Cluster 1:* Predominantly patients with local residual disease, low rates of gross total resection (GTR), and limited radiotherapy. Mortality rate: 63%. This group had the worst overall prognosis.
- *Cluster 2:* Enriched in glioma cases, high rates of radiotherapy, longer symptom duration, and mostly supratentorial localization. Mortality rate: 43%. This group had more heterogeneous outcomes.
- *Cluster 3:* Composed mainly of embryonal tumors, with high rates of GTR and aggressive multimodal treatment. Mortality rate: 39%. Despite tumor aggressiveness, outcomes were favorable due to intensive therapy.

A chi-squared test on `Death` distribution across clusters yielded a significant p-value ($p < 0.05$), indicating that mortality is not uniformly distributed across groups. Standardized residuals showed that Cluster 1 contributed most to the deviation from independence.

Unsupervised clustering on the original dataset did not reveal meaningful patient groupings. However, when applied to the UMAP-reduced space, clustering uncovered three groups with clinically interpretable profiles and significantly different outcomes. These findings suggest that nonlinear dimensionality reduction helps expose latent structure relevant to treatment intensity, tumor type, and prognosis in pediatric brain tumors.

7. Survival Analysis

To investigate survival outcomes and prognostic indicators in pediatric brain tumors, we employed Kaplan-Meier estimation and multivariate Cox regression approaches. The survival endpoint was modeled using the binary variable `Death` and `OS` (overall survival time).

Kaplan-Meier Estimates

We first computed the overall survival curve using the Kaplan-Meier estimator. As shown in Figure 1, the survival distribution reflects a

steep decline in the first 20 months, followed by a plateau suggestive of long-term survivors.

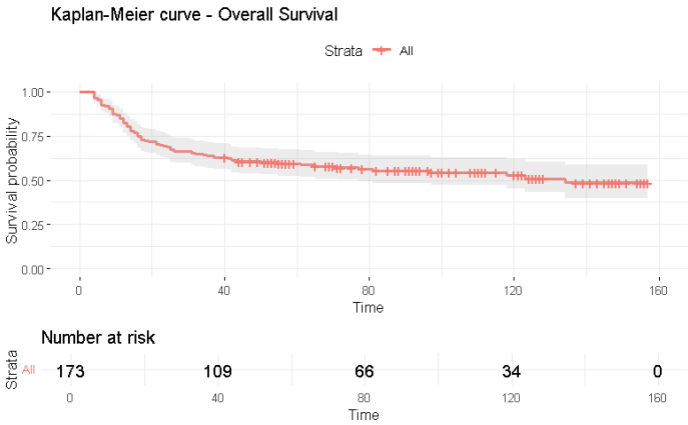

Figure 1. Kaplan-Meier survival curve for the full patient cohort. Shaded areas represent 95% confidence intervals.

Kaplan-Meier curves stratified by clinical variables, visible in Appendix 1, revealed several significant patterns:

- *Extent of Disease*: Survival significantly differed across categories. Patients with disseminated disease (DD) exhibited the poorest prognosis, while those with no evidence of disease (NED) had the most favorable outcomes ($p = 0.00014$).
- *Operation Type*: Gross Total Resection (GTR) was strongly associated with improved survival compared to non-radical interventions ($p = 0.00023$).
- *Tumor Histology*: High-Grade Gliomas (HGG) and tumors of unknown histology showed sharply decreased survival rates ($p < 0.0001$).
- *Radiotherapy*: Although not statistically significant ($p = 0.076$), patients receiving radiotherapy exhibited a visual trend toward better survival.
- *Adjuvant Chemotherapy and Sex*: No significant differences in survival were detected.

Cox Proportional Hazards Models

We initially fitted a full Cox proportional hazards model including all variables from the dataset. As expected, many covariates were not statistically significant. We then built a more parsimonious multivariate Cox model including nine clinically and statistically relevant predictors:

```
Sex, HGG, Unknown, Radiotherapy,
Operation_type_bin, Extent_of_disease3,
Adjuvant_HT, GCT___NGCT,
Duration_of_sympthoms_before_diagnosis_days
```

The model showed excellent discriminative ability with a concordance index (C-index) of 0.813 and good overall fit (AIC = 666). Significant negative prognostic factors included unknown histology, high-grade glioma (HGG), disseminated disease, and lack of gross total resection. Radiotherapy appeared as a strong protective factor.

Proportional Hazards Assumption The proportional hazards assumption was tested using `cox.zph`. Global p-value indicated a violation ($p = 0.02$), particularly due to the variables `Adjuvant_HT` (systematic deviation in residuals) and `Unknown` (initial deviation outside confidence bands).

Stratified Cox Regression

To address these violations, we first estimated a stratified Cox model using `Adjuvant_HT` as the stratification variable. The model passed the proportionality test globally ($p = 0.48$), though a mild violation remained for `Unknown` ($p = 0.034$).

To fully correct for this, we estimated a final stratified model on both `Adjuvant_HT` and `Unknown`. The global proportional hazards test confirmed that all covariates met the assumption (GLOBAL $p = 0.95$). The model achieved a C-index of 0.773, indicating good discriminative performance.

Interpretation of Results

The final model identified four independent, strong and statistically significant predictors of mortality:

- **High-Grade Glioma (HGG)**: HR = 9.18 [95% CI: 4.78 – 17.64]
- **Disseminated Disease (Extent_of_disease3)**: HR = 3.37 [95% CI: 1.63 – 6.97]
- **No Gross Total Resection (Operation_type_bin = 0)**: HR = 0.33 [95% CI: 0.18 – 0.60]
- **Radiotherapy**: HR = 0.17 [95% CI: 0.06 – 0.48]

Other covariates, such as `GCT___NGCT`, showed hazard ratios below 1 (HR = 0.16) but were not statistically significant due to wide confidence intervals. Variables such as `Sex` and `Duration_of_sympthoms_before_diagnosis_days` had no relevant effect on survival.

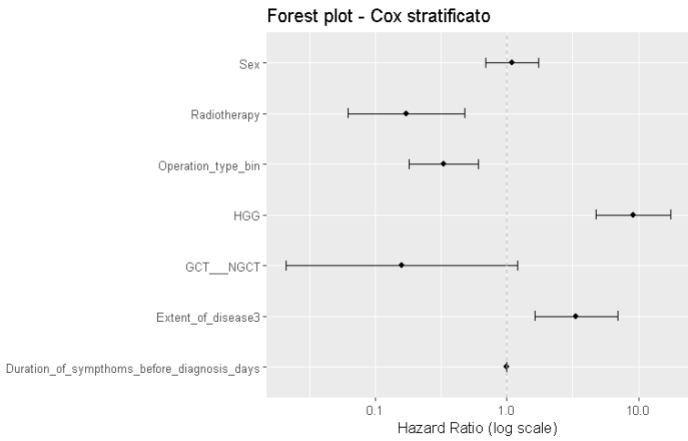

Figure 2. Forest plot of hazard ratios from the stratified Cox model (log scale).

The model results support the importance of tumor biology, disease extent, and treatment intensity in determining prognosis in pediatric brain tumors. Stratification was effective in resolving proportionality violations, and the final model is robust and interpretable.

8. Supervised Learning for Mortality Prediction

To evaluate the predictive capacity of clinical variables on patient mortality, we trained a set of supervised classification models using the binary outcome `Death`. The dataset was pre-processed to exclude the `OS` variable to avoid leakage, and all predictors were standardized where required.

Model Training and Evaluation

Five algorithms were considered: Random Forest, XGBoost, Support Vector Machine (SVM), k-Nearest Neighbors (KNN), and a feedforward Neural Network. Models were trained using a 10-fold cross-validation strategy, and performance was evaluated on a hold-out test set (20% of the data).

For the Neural Network, features were normalized via centering and scaling. The target variable was encoded as a factor with two levels: `No` and `Yes`.

ROC Curve Comparison

The ROC curves obtained on the test set using class probability thresholds fixed at 0.5. Among all models, Random Forest and XGBoost achieved the highest AUC values (0.78 and 0.77, respectively), followed by Neural Network (AUC = 0.72). KNN demonstrated the weakest performance with an AUC of 0.55 and poor specificity.

Threshold Optimization

To improve predictive balance, we recalculated predictions using the optimal thresholds derived from the Youden Index on ROC curves.

Threshold optimization led to improved model calibration. The SVM achieved the best overall results (AUC = 0.70, Accuracy = 82%, Balanced Accuracy = 0.82), followed by Random Forest and XGBoost (AUC ≈ 0.83 and 0.82, respectively). The Neural Network achieved an AUC of 0.79 with excellent specificity (0.90). KNN remained suboptimal (AUC = 0.53) and exhibited poor class separation.

Feature Ranking

To interpret the models, we computed variable importance scores across all algorithms.

Across models, a consistent subset of predictors emerged as highly relevant:

- **HGG**: High-grade glioma status was the strongest risk factor, reflecting tumor aggressiveness.
- **Operation_type_bin**: Extent of surgery (GTR vs non-GTR) showed high predictive value.
- **Unknown Histology**: Absence of a definitive diagnosis was associated with worse prognosis.
- **Duration of Symptoms**: Longer pre-diagnosis symptom duration correlated with increased mortality.

Other models such as SVM and Neural Network provided flatter rankings. Notably, the Neural Network occasionally overemphasized less relevant variables such as `Sex`.

Summary of Model Performance

Table 1. Model performance metrics on test set with optimal threshold.

Model	AUC	Accuracy	Sensitivity	Specificity
Random Forest	0.83	0.76	0.64	0.85
XGBoost	0.82	0.76	0.64	0.85
SVM	0.70	0.82	0.79	0.85
Neural Network	0.79	0.76	0.57	0.90
KNN	0.53	0.41	1.00	0.00

Overall, Random Forest and SVM emerged as the most balanced and effective classifiers, while KNN showed clear signs of overfitting and limited generalization capacity. Across models, feature importance rankings and ROC analysis consistently identified tumor biology (e.g., HGG), diagnosis certainty (Unknown histology), and surgical intervention (Operation type) as the strongest determinants of mortality risk.

Among all models, Random Forest stood out as the most robust and interpretable. It achieved the highest AUC (0.83), maintained a balanced trade-off between sensitivity (0.64) and specificity (0.85), and provided clinically coherent variable rankings. These findings suggest that Random Forest offers the best compromise between predictive accuracy and model transparency, making it a reliable choice for mortality prediction in this pediatric oncology cohort.

9. Conclusions

Descriptive and Survival Analysis

Descriptive and survival analyses revealed strong associations between mortality and several clinical features. High-grade gliomas (HGG), disseminated disease at diagnosis, unknown histology, and lack of gross total resection were consistently associated with reduced survival. Conversely, radiotherapy and radical surgery showed protective effects.

These findings were confirmed through Kaplan–Meier survival curves and Cox proportional hazards models. The final stratified Cox model, adjusted for violations of the proportional hazards assumption, achieved a concordance index of 0.77. The model effectively identified key prognostic factors and supported clinically meaningful risk stratification based on tumor biology, disease extent, and treatment intensity.

Unsupervised Learning and Clustering

Unsupervised learning techniques—particularly dimensionality reduction with UMAP followed by clustering—identified three distinct patient profiles. These clusters showed meaningful differences in clinical characteristics and survival outcomes. Although clustering in the raw space produced limited structure, projection into the UMAP space allowed more coherent patient grouping. A chi-squared test confirmed statistically significant variation in mortality across clusters.

Supervised Classification

Supervised machine learning models were employed to predict mortality. Random Forest and XGBoost yielded the best overall performance, with AUC values of 0.83 and 0.82 respectively, and balanced accuracy above 0.74. SVM also performed well after threshold optimization, while KNN displayed overfitting and poor specificity.

Feature importance rankings consistently identified HGG, unknown histology, surgical intervention type, and symptom duration as the most relevant predictors. Random Forest, in particular, combined high predictive power with clinically interpretable variable importance, confirming its value in risk prediction.

Limitations

Several limitations should be noted. First, the dataset is single-center and limited in size (173 patients), potentially reducing generalizability and increasing the risk of overfitting. Second, the retrospective nature of the data may introduce residual confounding due to unmeasured variables. Third, the presence of "unknown" histologies—often reflecting non-biopsied or diagnostically uncertain cases—limits clinical interpretability and may bias model inference.

Moreover, missing data were handled through statistical imputation, which may introduce artifacts if the missingness mechanism is not random. Finally, treatment details (e.g., chemotherapy regimens, RT dosage) were simplified, potentially masking relevant variability in care.

References

[1] D. Stanić, D. Grujičić, T. Pekmezović, *et al.*, "Clinical profile, treatment and outcome of pediatric brain tumors in serbia in a 10-year period: A national referral institution experience", *PLOS ONE*, vol. 16, no. 10, e0259095, 2021. DOI: 10.1371/journal.pone.0259095.

A. Kaplan-Meier Survival Curves by Variable

This appendix provides Kaplan-Meier survival plots stratified by key clinical variables discussed in Section 7. Shaded regions denote 95% confidence intervals, and *p*-values are based on the log-rank test.

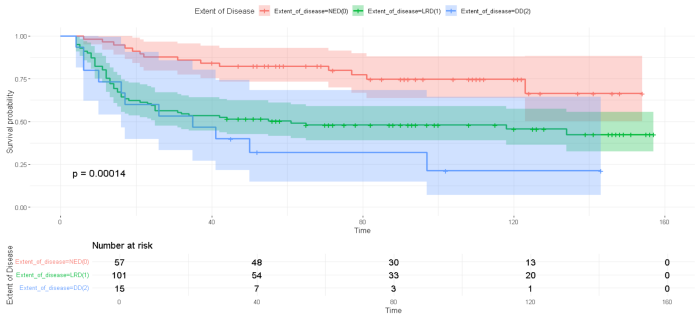

Figure 3. Kaplan-Meier survival curves stratified by Extent of Disease (NED, LRD, DD).

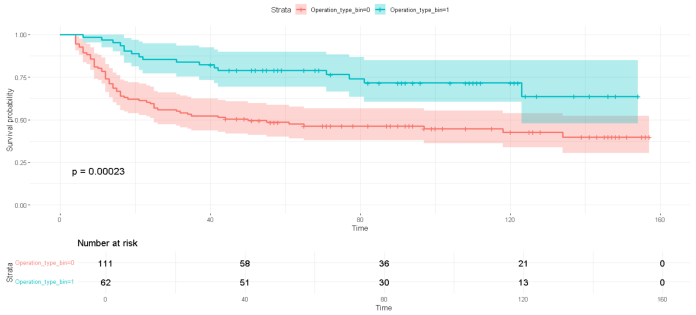

Figure 4. Kaplan-Meier survival curves by surgical intervention (GTR vs. non-GTR).

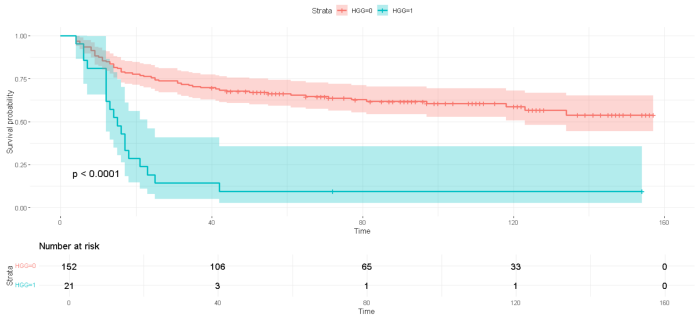

Figure 5. Kaplan-Meier survival curves for patients with and without High-Grade Glioma (HGG).

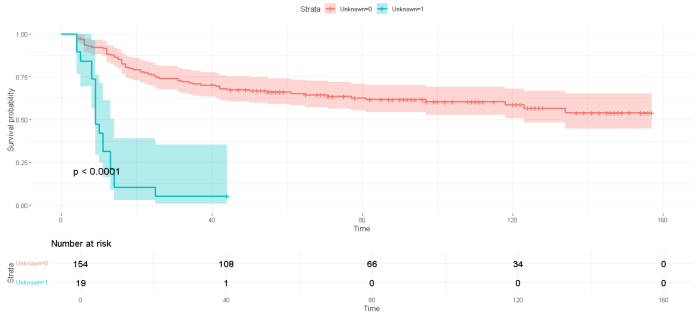

Figure 6. Kaplan-Meier survival curves for patients with unknown histological diagnosis.

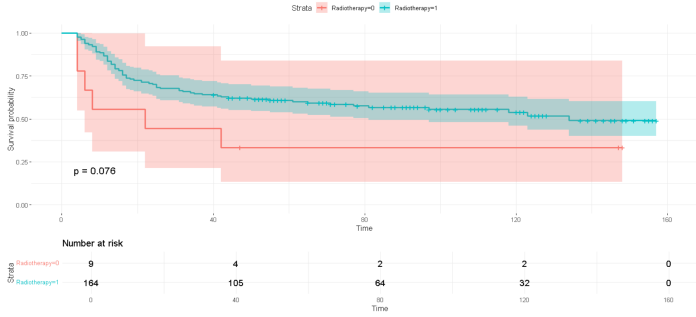

Figure 7. Kaplan-Meier survival curves by radiotherapy status.

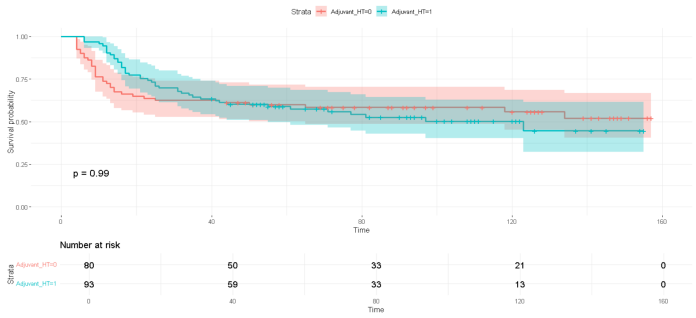

Figure 8. Kaplan-Meier survival curves by adjuvant chemotherapy administration.

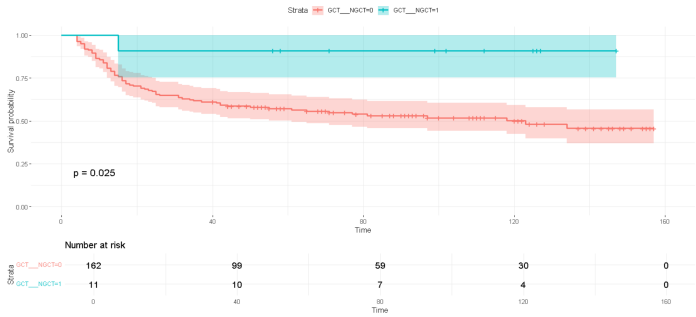

Figure 9. Kaplan-Meier survival curves for patients with GCT/NGCT histology.